



Supporting Information for:

***In utero* ultrafine particulate matter exposure causes offspring pulmonary immunosuppression**

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Supplementary text

Figs. S1 to S3 (Figs. S1 through S3 are referenced in the main manuscript)

Tables S1 and S2 (Tables S1 is referenced in the main manuscript)

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Supplementary text

UFP Generation and Mouse Exposure Model

Mice were housed in a climate-controlled room with 12/12h light/dark cycle. Females received 9% fat breeder diet (Harlan, Indianapolis, IN), and males received standard rodent chow (4%) (LabDiet, St. Louis, MO). Food and water were provided *ad libitum*. Female C57Bl/6 ($n = 23$) and BALB/c ($n = 23$) mice were time-mated, and the presence of a vaginal plug defined gestational day (GD) 0. Beginning on GD0, dams were randomized and placed into whole body exposure chambers wherein they were exposed to either UFPs ($n = 11$) or FA ($n = 12$) from 8:00 am to 2:00 pm (6 hours) daily through GD18. Exposure chambers consisted of a 12" x 8" x 32" stainless steel box with separated inner compartments and a 1/4" clear cast acrylic lid (Figs. S1 and S2). Air was continuously pumped through the chamber by stainless steel aerosol distribution lines attached to the lid and out return lines on the bottom. UFPs were generated utilizing a commercial constant output atomizer (TSI 3076, TSI, Inc., Shoreview, MN). Particles were then conditioned, including adjustment of particle number concentration in a dilutor, water vapor removal in a multi-tube Nafion drier, organic vapor removal in a denuder filled with Spectrum XB-17 reactive adsorbent, and charge neutralization by a Po-210 bipolar diffusion charger. The concentration and size distribution of UFPs in the chambers were constantly monitored with a Scanning Mobility Particle Sizer (SMPS) system and adjusted by changing the corresponding UFP dilution ratios at the source. The SMPS operated with a sheath flow of 6.5 liters per minute and a sample flow of 1 liter per minute. The mass concentration of accumulation mode PM was maintained near $100 \mu\text{g}/\text{m}^3$, corresponding to a total number concentration of about 10^5 particles/ cm^3 for 50 nm geometric mean diameter particles. The particle mass concentration was calculated from the measured size distribution and particle density, which was estimated from the weighted average density of 10 wt % diesel exhaust, 44 wt % NH_4NO_3 , 39 wt % $(\text{NH}_4)_2\text{SO}_4$, and 7 wt % KCl, with a corresponding value of 1.63 g cm^{-3} . The detailed procedures for generating and quantifying UFPs have been described elsewhere (1, 2). Following exposure on GD18, mice were removed to individual housing and allowed to deliver spontaneously. In the C57Bl/6 strain, 78% of dams ($n = 18$) were positively pregnant and sustained a litter, whereas fewer of the BALB/c mice, 52% ($n = 12$), successfully completed pregnancy and sustained a litter. Treatment did not have an effect on pregnancy outcomes. In the C57Bl/6 strain, 81% of dams exposed to UFPs ($n = 9$) and 75% of dams exposed to FA ($n = 9$) sustained a litter. In the BALB/c strain, 55% of dams exposed to UFPs ($n = 6$) and 50% of dams exposed to FA ($n = 6$) sustained a litter. Litters were randomly allocated to offspring experimental (allergen-exposed) or control groups equally, yielding 4-5 litters per group in the BALB/c strain and 3 litters per group in the C57Bl/6 strain.

Offspring Allergen Challenge

To determine offspring response to allergen challenge following *in utero* UFP exposure, we followed a well-characterized protocol entailing chronic dosing with house dust mite (HDM) (3). Starting on postnatal day (PND) 3, pups were briefly anesthetized with 3% isoflurane in oxygen and exposed to

either 10 μ g HDM in 10 μ L PBS or 10 μ L PBS alone intranasally (5 μ L in each nostril 30 seconds apart) on PNDs 3, 5, 7, 10, 12, and 14. On PNDs 17, 19, 21, 24, 26, and 28, pups were again briefly anesthetized and exposed to either 15 μ g HDM in 15 μ L PBS or 15 μ L PBS alone intranasally (7.5 μ L in each nostril 30 seconds apart). Beginning on PND3, mice were weighed and measured using a digital caliper (crown to rump) on each exposure day while still under light anesthesia. Offspring were sacrificed on PND31 (72 hours following the final allergen challenge) using an overdose of sodium pentobarbital.

Pulmonary Inflammation Determination

Bronchoalveolar lavage (BAL) fluid was collected in K₂EDTA tubes by tracheal cannulation and washing the lungs with 1.0 ml sterile PBS. BAL was kept on ice until analysis of airway inflammation. The left lung was then excised and inflated to a constant pressure of 25 cm with 10% formalin. The right lung was immediately frozen in liquid nitrogen and stored at -80°C. BAL samples were analyzed at the Texas Veterinary Medical Diagnostic Laboratory (TVMDL). Total number of cells in BAL fluid was counted with a hemocytometer. Differential counts of leukocytes were performed using cytopspin smear slides by clinical pathologists blinded to treatment group. Cells were classified as lymphocytes, macrophages, eosinophils, or neutrophils based on standard morphology. Fixed lungs were processed for routine histopathology. Sections were stained with hematoxylin and eosin (H&E) for histological evaluation and periodic-acid-Schiff (PAS) to measure number of goblet cells. Histological evaluation was performed using a scoring system ranging from 0-3 (0-no lesions; 1-mild; 2-moderate; 3-marked) in each of the following categories for a total possible score of 12: lymphoid hyperplasia, peribronchiolar eosinophils, perivascular eosinophils, and intraalveolar eosinophils. Number of goblet cells were assessed in 20 different 10x fields using a scoring system ranging from 0-4 (0- no goblet cells; grade 1, <25%; grade 2, 25-50%; grade 3, 51-75%; grade 4, >75%) corresponding to the percentage of goblet cells in the bronchiolar epithelial cells, as previously described (4). Mean scores of the number of goblet cells in each mouse were calculated. All histological assessments were carried out by veterinary pathologists blinded to treatment groups.

Systemic Inflammation Determination

Offspring blood was collected into K₂EDTA microtainer tubes (Becton Dickson, New Jersey), separated within 8 hours of collection and stored at -80° C until analysis. Circulating cytokines and chemokines were analyzed using a Milliplex MAP Mouse Cytokine/Chemokine Kit from Millipore (Millipore Corporation, Billerica, MA). The following cytokines were analyzed: eotaxin, G-CSF, GM-CSF, IFN, IL-1 α , IL-1 β , IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-9, IL-10, IL-12 (p40), IL-12 (p70), IL-13, IL-15, IL-17, IP-10, LIF, LIX, MCP-1, M-CSF, MIG, MIP-1 α , MIP-1 β , MIP-2, RANTES, TNF α , and VEGF.

Quantitative real-time PCR

Total RNA was isolated from lung tissue using RNeasy Mini Kit (Qiagen) and converted to cDNA using iScript cDNA Synthesis Kit (Bio-Rad, Hercules, CA). cDNA was amplified and quantified using gene-specific primers (Table S2). Quantitative real-time PCR was performed CFX384 Touch™ Real-Time

PCR Detection System (Bio-Rad, Hercules, CA) with SYBR Green I Master (Roche, Indianapolis, IN). Relative expression was calculated using delta-Ct method normalized to reference gene HPRT.

Immunohistochemistry

To determine number of activated T lymphocytes, histological sections were used for immunohistochemistry using an anti-FOXP3 antibody. After manual deparaffinization and rehydration of histological slides, antigen retrieval was performed on a pressure cooker utilizing the commercial compound Diva decloaker (Biocare, Pacheco, CA) for 20 min under 121°C. Sections were treated with 0.3% hydrogen peroxide in order to block endogenous peroxidase activity for 45 minutes, and with a protein blocker for endogenous protein activity (Thermo Fisher Scientific, Waltham, MA) for 10 minutes. Slides were incubated with an anti-FOXP3 antibody (ab20034; Abcam, Cambridge, UK) overnight at 4°C. On the next day, slides were incubated with a goat anti-mouse/rabbit polymer visualization system (Thermo Fisher Scientific, Waltham, MA) for 10 minutes, and color development was achieved using 3, 3'-Diaminobenzidine (DAB, Thermo Fisher Scientific) as the chromogen. Slides were counter-stained with hematoxylin and examined under light microscopy. Activated T cells had intracytoplasmic immunolabeling, and total number of positive cells were counted in 10 400X fields.

Statistical Analysis

Statistical analysis was run using SigmaPlot statistical software (Systat Software, Inc., San Jose, California). Groups (FA-PBS, FA-HDM, PM-PBS, and PM-HDM) were compared using analysis of variance (ANOVA) tests except when data did not meet normality criteria and a non-parametric version, Kruskal-Wallis, test was applied. Likewise, we compared groups using the Mann-Whitney rank sum test, and the significance was indicated in the figure legends.

Materials

Diesel Particulate Matter Standard Reference Material (SRM 2975) was purchased from the National Institute of Standards and Technology (Gaithersburg, MD). House Dust Mite (*Dermatophagoides farina*, Lot 220744) greater than 99% purity was purchased from GREER (Lenoir, NC). Phosphate Buffered Saline was purchased from Sigma Chemical Co. (St. Louis, MO). All other chemicals and reagents used were obtained commercially at the highest available purity. Male and female C57Bl/6J and BALB/cJ 8-week-old mice were purchased from The Jackson Laboratory (Bar Harbor, ME). The experimental protocol was reviewed and approved by the Institutional Animal Care and Use Committee, Texas A&M University.

Table S1. Changes in airway inflammation and mucus production following *in utero* exposure to filtered air or PM and offspring allergen challenge in C57Bl/6 and BALB/c mice.

Mouse Strain	Group	LF	PB Eos	PV Eos	A Eos	Total	PAS
C57Bl/6	FA PBS	0 ± 0	0.1 ± 0.4	0 ± 0	0 ± 0	0.1 ± 0.4	0.4 ± 0.2
	FA HDM	1.3 ± 0.7 ^a	1.9 ± 0.3 ^a	2.1 ± 0.8 ^a	0.7 ± 0.7	6.0 ± 1.8 ^a	1.8 ± 0.2 ^a
	PM PBS	0.1 ± 0.3	0 ± 0	0 ± 0	0.1 ± 0.3	0.2 ± 0.6	0.3 ± 0.2
	PM HDM	1.1 ± 0.5 ^b	1.5 ± 0.8 ^b	1.4 ± 0.8 ^b	0.4 ± 0.8	4.5 ± 2.1 ^{b,c}	1.6 ± 0.4 ^b
	Absolute Control	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0.2 ± 0
BALB/c	FA PBS	0.6 ± 0.5	0.3 ± 0.5	0 ± 0	0.3 ± 0.5	1.1 ± 1.0	0.6 ± 0.5
	FA HDM	0.8 ± 1.2	1.2 ± 1.6	0.7 ± 1.2	0.3 ± 0.8	3.0 ± 0.8	1.8 ± 0.6 ^a
	PM PBS	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0.2 ± 0.1
	PM HDM	1.1 ± 0.9 ^b	1.5 ± 1.4 ^b	1.2 ± 1.1 ^b	0.5 ± 0.8	4.3 ± 3.5 ^b	2.0 ± 0.3 ^b

Histological evaluation was performed on H&E stained slides using a scoring system ranging from 0-3 (0-no lesions; 1-mild; 2-moderate; 3-marked) in each of the following categories: lymphoid follicular hyperplasia (LF), peribronchiolar eosinophils (PB Eos), perivascular eosinophils (PV Eos), and intraalveolar eosinophils (A Eos). The number of goblet cells were assessed using a scoring system ranging from 0-4 (0-no goblet cells; grade 1, <25%; grade 2, 25-50%; grade 3, 51-75%; grade 4, >75%) corresponding to the percentage of goblet cells in the bronchiolar epithelial cells in PAS stained slides. Average scores, shown above ± SD, were compared among groups in C57Bl/6 mice: ^ap<0.001 (FA HDM vs. FA PBS); ^bp<0.001 (PM HDM vs. PM PBS); ^cp=0.048 (FA HDM vs PM HDM); BALB/c mice: ^ap<0.05 (FA HDM vs. FA PBS); ^bp<0.005 (PM HDM vs. PM PBS).

Table S2. Primers used to amplify cDNA.

	Forward primer	Reverse Primer
Muc5ac	5'-CCCATTCCTGGGTCCTAGC-3'	5'-GATTTTCAGAGAGTCCTGTTGAGC-3'
Muc5b	5'-CATGGATGGCTGCTTCTGT-3'	5'-TAAATTCGGCCACCGTGT-3'
IFN γ	5'-GCGTCATTGAATCACACCTGA-3'	5'-CTGGACCTGTGGGTTGTTGA-3'
IL4	5'-CCATATCCACGGATGCGACA-3'	5'-GACGAGCTCACTCTCTGTGG-3'
IL5	5'-GAGGCTTCCTGTCCCTACTC-3'	5'-CCCACGGACAGTTTGATTCTT-3'
IL13	5'-CCTCTGACCCTTAAGGAGCTTAT-3'	5'-CGTTGCACAGGGGAGTCT-3'
IL17	5'-AGACTACCTCAACCGTTCCAC-3'	5'-CACTGAGCTTCCCAGATCACA-3'
IL10	5'-AGAGCCACATGCTCCTAGA -3'	5'-GTCCAGCTGGTCCTTTGTTT-3'
TGF- β	5'-CACCAAAGTCCTCAGCCTGT-3'	5'-GCTGTTCGATCTTGGGCGTA-3'
HPRT	5'-TCCTCCTCAGACCGCTTTT-3'	5'-CCTGGTTCATCATCGCTAATC-3'

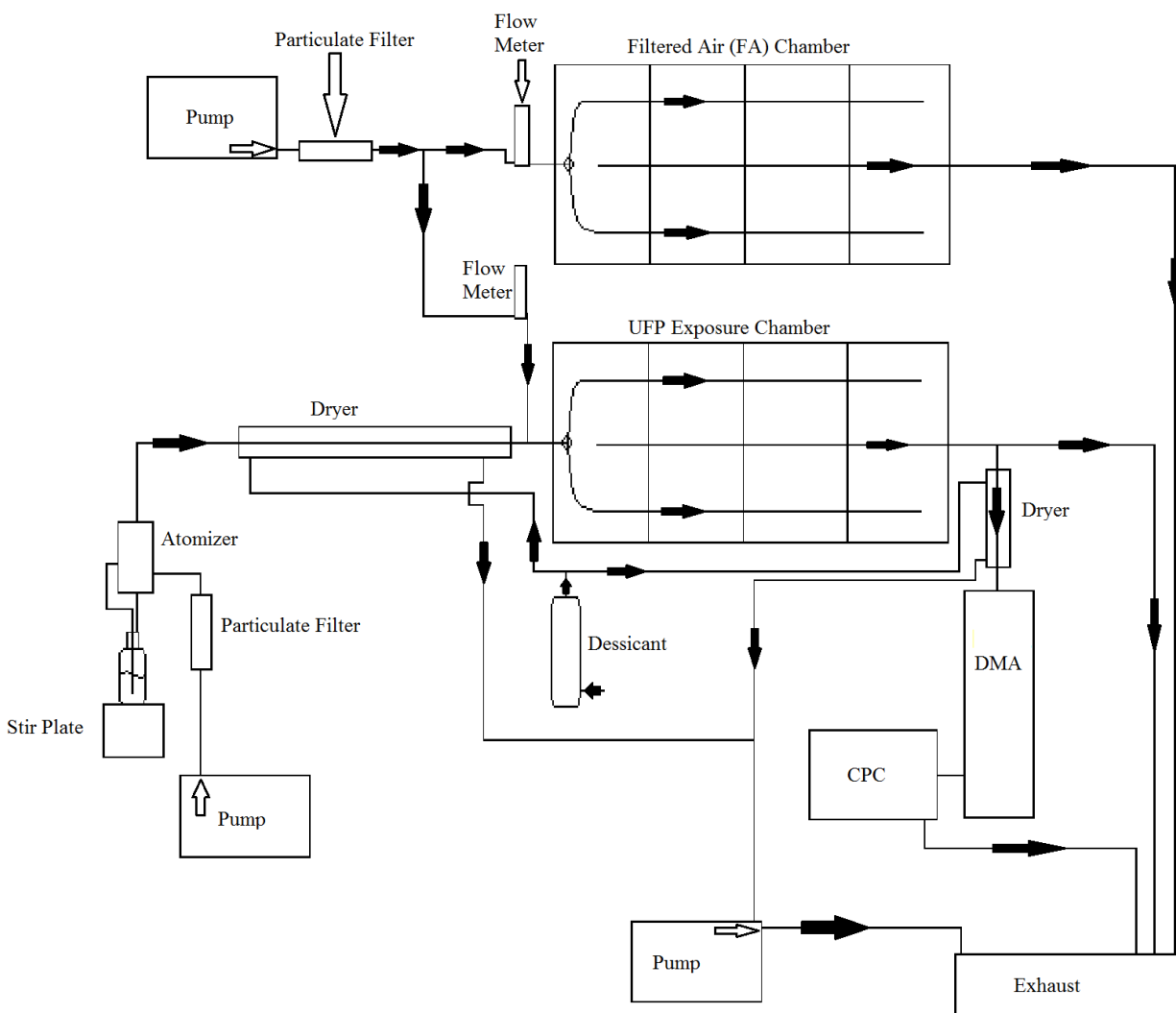


Fig. S1. Flow chart for the animal exposure experiment. CPC denotes condensation particle counter, and DMA denotes differential mobility analyzer. In the clean chamber, room air passed through a particulate filter to remove particles and flowed through the chamber with a controlled flow rate. For the polluted chamber, particles were produced by the atomizer and dried prior to introducing to the chamber. The particle laden flow was then diluted with clean air and introduced to the chamber. The particle size distribution was quantified after the chamber by the DMA.

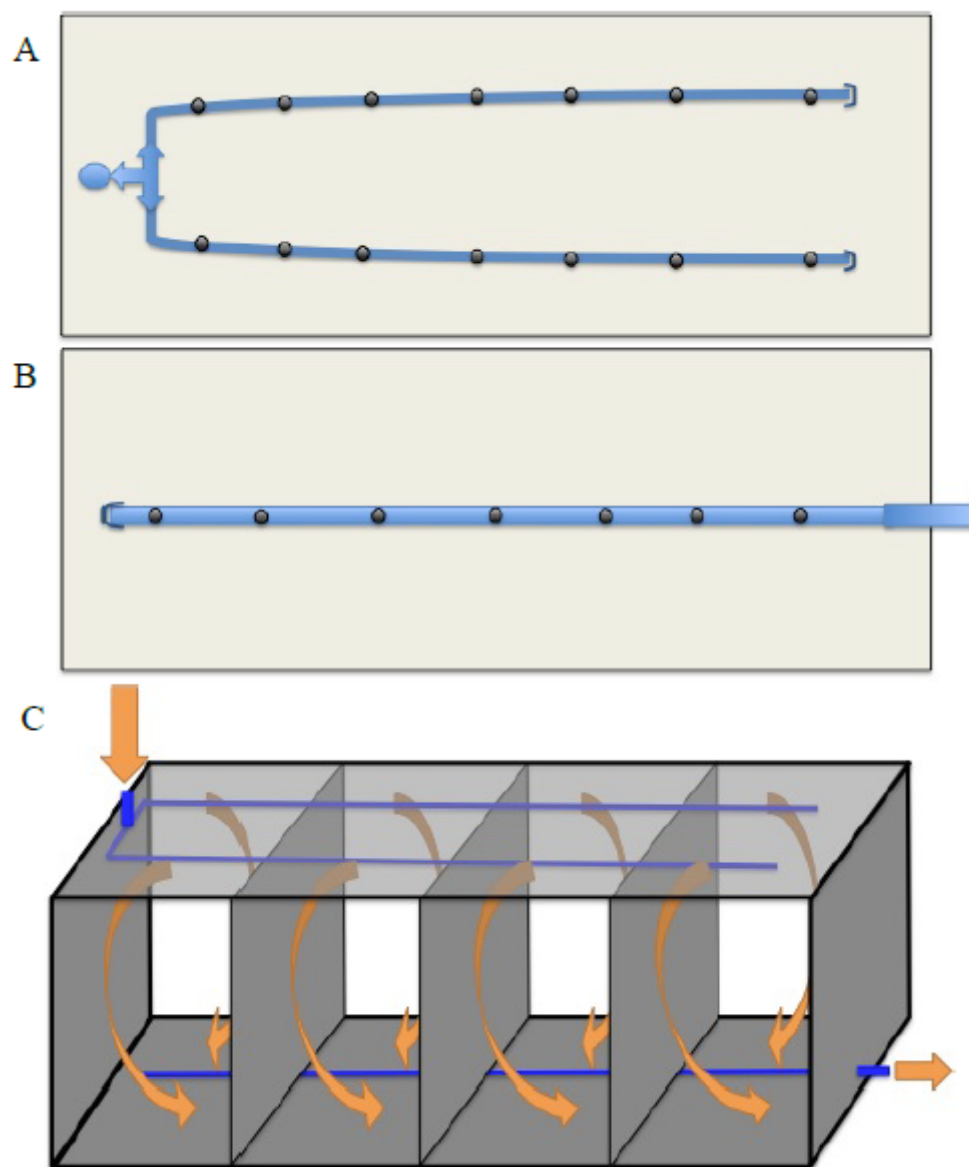


Fig. S2. Schematic of the animal exposure chamber. (A) Top view of the chamber for introducing particles. (B) The bottom view of the chamber for exhaust. (C) The flow patterns inside the chamber.

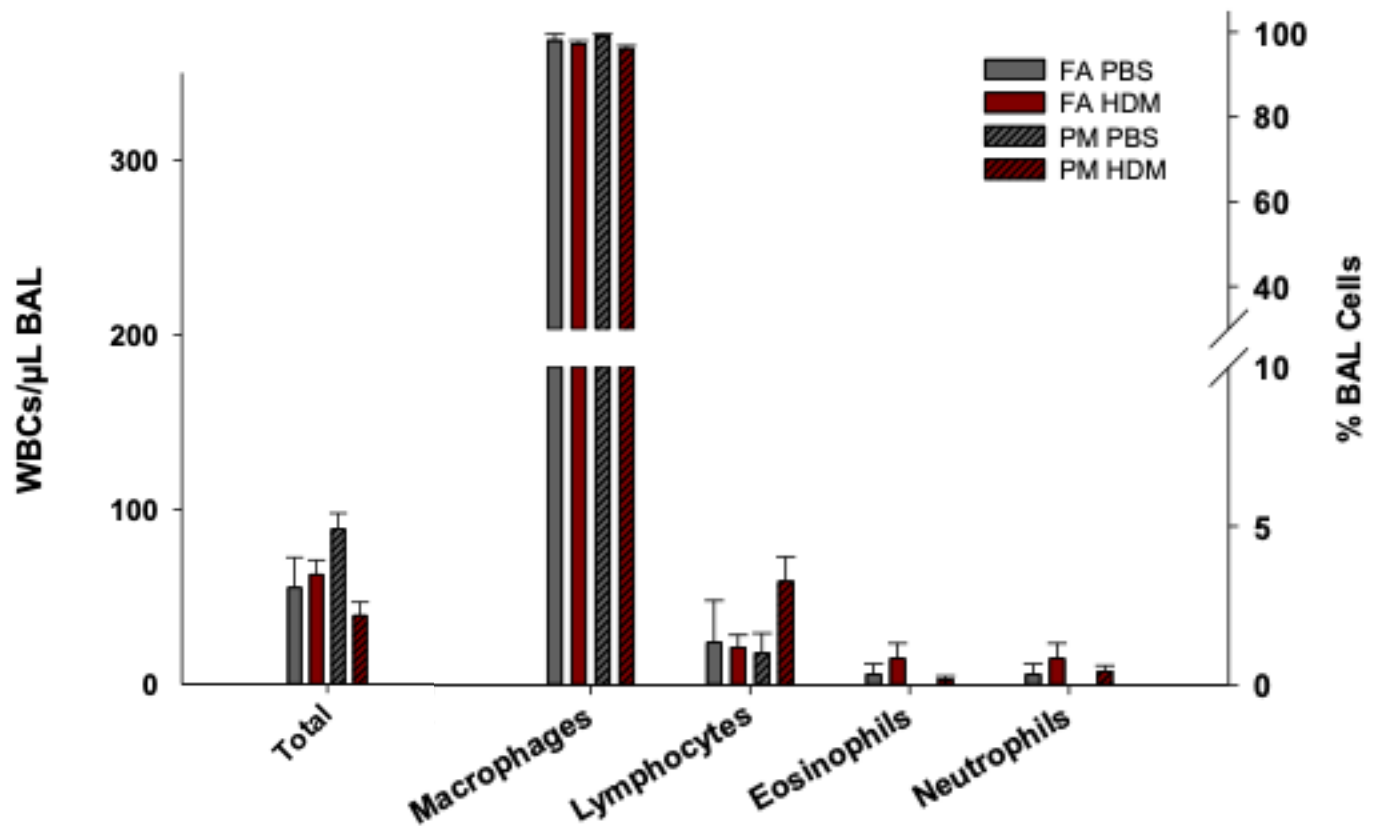


Fig. S3. White blood cells (WBCs) per μL in BAL expressed as mean $\pm$ SE (left axis) and the distribution of macrophages, lymphocytes, eosinophils and neutrophils in BAL expressed as percentage (right axis). C57Bl/6 offspring pooled from 4-5 litters per group, FA-PBS ($n = 3$), FA-HDM ($n = 6$), PM-PBS ($n = 5$), PM-HDM ($n = 12$).

References

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